Constructive and Destructive Training Regimes: A Framework for Interpreting SGD Stability

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Parameter updates can reinforce or cancel one another even when training loss continues to decrease. An exact finite-time decomposition of stochastic gradient descent (SGD) displacement separates gradient magnitudes from interactions across all temporal lags. Consecutive-gradient cosine provides a local directional diagnostic, while normalized displacement measures net motion relative to traveled distance. Pairing adjacent updates separates immediate cancellation from accumulation of the residual motion. Local expansions connect these quantities to curvature and sampling noise without identifying alignment sign with stability or objective progress. Full-batch experiments exhibit gradient reversal alongside continued loss reduction. Matched mini-batch continuations show that sampling changes both gradient estimates and the paths traversed through the objective. Under Adam, raw-gradient diagnostics remain informative about the loss landscape traversed by adaptive steps. Controlled full-batch experiments reveal loss spikes associated with pre-rise gradient reversal and norm amplification. Loss-defined events quantify their magnitude and recovery, while descriptive regressions relate the initial regime and subsequent norm growth to relative peak loss. The resulting framework supports the formulation and evaluation of informed learning-rate interventions; it does not prescribe an alignment-maximizing schedule.**

***Index Terms*—Stochastic Gradient Descent, Adam, Gradient Alignment, Parameter Displacement, Learning-Rate Scheduling**

I. Introduction

Neural scaling laws have ushered in new paradigms for interpreting deep learning dynamics. By establishing that performance gains follow predictable behavior relative to model size, dataset scale, and compute, stochastic optimization methods have been pushed toward increasingly extended training horizons while operating under tight efficiency constraints [1]–[4]. In this regime, optimization stability becomes an important practical requirement, since unstable training may result on slow convergence and high sensitivity to hyperparameter choices. In practice, stability is often maintained through a set of heuristics, such as momentum, batch-size scaling, normalization, and learning-rate schedules [5]–[9]. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

Training becomes unstable when updates are too large relative to sharp directions of the loss landscape, so that they overshoot rather than converge towards a minimum [10], [11]. Stochastic extensions, such as the break-even point framework, argue that instability depends jointly on sharpness and curvature variance, particularly during an early phase in which learning rate and batch size strongly shape further trajectory [12]. In contrast, however, the Edge-of-Stability literature has shown that deep networks often continue to make progress near, or even beyond, the nominal sharpness threshold, suggesting that curvature alone is not a sufficient explanation [13]–[15]. These results explain how rapid oscillations can accompany longer-term descent. Studies of optimization trajectories also distinguish directionality at the trajectory and individual-update scales [16]. A complementary measurement question is how much of the traveled path cancels within neighboring updates, and how coherently the remaining displacements accumulate. Gradient information already computed during optimization makes this accounting possible without estimating the full Hessian spectrum or gradient-noise covariance at every update.

To address this question, we characterize constructive and destructive training regimes through consecutive-gradient alignment and gradient magnitude. Positive alignment, $\rho_t > 0$, identifies a constructive regime in which successive SGD steps reinforce one another, whereas negative alignment, $\rho_t < 0$, identifies a destructive regime in which successive steps partially reverse preceding parameter changes. We connect these regimes to directional curvature, κ_t, and obtain a gradient-based curvature diagnostic from consecutive gradient norms and their inner product, without additional Hessian computations. The formulation also makes the contribution of mini-batch sampling explicit, clarifying how noise affects the interpretation of the observed alignment.

Destructive alignment alone does not imply instability. Within a local quadratic approximation, reversal and gradient growth jointly determine whether an SGD step decreases or increases the loss. This relationship explains how destructive regimes can sustain oscillatory descent, while sufficiently amplified reversals can produce upward loss excursions. Reading alignment and gradient norms together therefore provides a geometric interpretation of signals already available during training.

We examine these relationships through controlled full-batch and mini-batch experiments across image-classification and language-modeling tasks, using multilayer perceptrons,

Thiago Mattar, and Yan G. Ferreira are with the Graduate Program in Electrical Engineering, Universidade Federal de Minas Gerais, Belo Horizonte 31270-901, Brazil (e-mail: thiagocmattar@gmail.com; yanvgferreira@gmail.com).

Cristiano L. Castro is with the Graduate Program in Electrical Engineering and the Department of Electrical Engineering, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Minas Gerais 31270-901, Brazil (e-mail: crislcastro@gmail.com).

Frederico Coelho, and Antonio P. Braga are with the Graduate Program in Electrical Engineering and the Department of Electronic Engineering, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Minas Gerais 35931-008, Brazil (e-mail: fredgfc@ufmg.br; apbraga@ufmg.br).

residual networks, and Transformers. The GD experiments isolate training dynamics from batch sampling, and the SGD continuations measure the combined effects of sampling and changing trajectories. The Adam experiments examine raw-gradient reversal and amplification along adaptively chosen updates. These experiments illustrate the proposed characterization and motivate controlled interventions, including learning-rate adjustment and gradient clipping.

In summary, our main contributions are:

- A geometric characterization of local reversal and cumulative displacement, including an exact paired-update factorization and a zero-cross-term reference, distinguished from objective progress.
- A connection between alignment, gradient magnitude, and directional curvature under SGD, yielding a curvature diagnostic from existing gradients without additional gradient evaluations or Hessian-vector products, with an explicit account of mini-batch noise.
- A local analysis of how gradient reversal and norm amplification jointly produce loss increases, clarifying the conditions separating oscillatory descent from upward loss excursions.
- Controlled experiments across multiple datasets and architectures that illustrate these relationships under SGD and examine the corresponding raw-gradient dynamics under Adam, providing a basis for informed training interventions.

The remainder of the paper is organized as follows. Section II reviews training-stability perspectives. Section III develops the displacement formulation and its diagnostics. Section IV examines full-batch GD and mini-batch sampling. Section V extends the interpretation to Adam and outlines the empirical requirements for informed scheduling. Section VI concludes.

II. Stability Analysis of SGD

Intuitively, smooth optimization occurs when updates reinforce each other and gradually shift parameters toward a minimum of the loss. Larger steps can overshoot, oscillate, and partially undo preceding parameter changes. Understanding when such behavior remains productive, and when it produces instability, requires examining how successive updates interact. Two factors are central to this analysis: the *local curvature* of the loss landscape and the *stochasticity* introduced by mini-batch sampling.

A. Local Curvature of the Loss Function and Stability

Training is a discrete-time dynamical system whose local response depends on curvature. A step that produces smooth descent in a shallow direction can overshoot in a sharper one. The Hessian quantifies this directional variation. Classical analyses [10], [11] connect its eigenvalues to the learning rate and local stability.

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t and g_t a mini-batch gradient evaluated at θ_{t-1}. For a constant learning rate $\eta > 0$, plain SGD follows

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

For full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$. With $H_t := \nabla^2\mathcal{L}(\theta_{t-1})$, the Jacobian of its update map is

$$J_t = I - \eta H_t. \tag{2}$$

On a fixed quadratic objective, a perturbation along a Hessian eigendirection is multiplied by $1 - \eta\lambda_i$ at each step. For a positive-definite Hessian, all components contract precisely when

$$\max_i |1 - \eta\lambda_i| < 1 \quad \Longleftrightarrow \quad \eta\lambda_{\max}(H_t) < 2. \tag{3}$$

For $0 < \eta\lambda_i < 1$, contraction preserves the component's sign. For $1 < \eta\lambda_i < 2$, its sign alternates while its magnitude shrinks. Above two, the component alternates and grows. These cases distinguish smooth approach, convergent oscillation, and instability of the fixed quadratic dynamics.

Neural-network Hessians are strongly anisotropic: a few outlier directions can be much sharper than the spectral bulk [17], [18]. Monitoring $\lambda_{\max}(H_t)$ therefore provides valuable stability information, but doing so throughout training is expensive. Even matrix-free eigensolvers require repeated Hessian–vector products beyond the ordinary gradient computation. An online diagnostic that recovers curvature information from those existing gradients would make this geometric analysis accessible during routine training.

B. Mini-Batch Stochasticity

Mini-batch SGD adds a sampling-dependent component to the full-gradient direction [19]–[21]. For independent uniform batches of b examples sampled without replacement from $N > 1$ examples,

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &= \frac{N-b}{b(N-1)}\Sigma(\theta_{t-1}). \end{aligned} \tag{4}$$

Here Σ is the covariance of per-example gradients. Writing $h_i := \nabla\ell_i(\theta_{t-1})$ and $\mu := N^{-1}\sum_i h_i = \nabla\mathcal{L}(\theta_{t-1})$,

$$\Sigma(\theta_{t-1}) := \frac{1}{N}\sum_{i=1}^{N}(h_i - \mu)(h_i - \mu)^\top. \tag{5}$$

Its trace measures gradient variability, while its directional structure describes how sampling perturbs the trajectory. Exact evaluation requires per-example information absent from the aggregated training gradient. For an inexpensive online diagnostic, the practical question is how this sampling variability changes the alignment of the gradients already being computed.

C. Curvature, Noise and the Edge of Stability

Learning rate and batch size also shape the geometry encountered during training. The break-even framework of Jastrzebski et al. [12] shows how their early values influence later curvature and gradient variability. This motivates examining curvature and sampling together along the trajectory.

The Edge-of-Stability (EoS) phenomenon makes the distinction between local instability and sustained convergence

especially clear. Full-batch GD often sharpens progressively until $\lambda_{\max}(H_t)$ fluctuates near or above $2/\eta$, while loss continues to decrease over longer timescales [13], [14]. A fixed quadratic with the same sharpness would be unstable, but a neural network changes its curvature as it moves. Self-stabilization analyses explain how growth along a sharp direction can induce a reduction in curvature, limiting the oscillation [15]. More recently, central flows describe the time-averaged trajectory together with the covariance of its rapid oscillations [22].

These results explain how short-term reversals can accompany longer-term progress. They also make the interaction between successive updates central to understanding convergence: the trajectory can repeatedly reverse in sharp directions while continuing to descend elsewhere.

D. Practical Stability Analysis in Large-Model Pretraining

Practitioners address spikes through monitoring, recovery, and preventive changes to the training recipe. PaLM recovered from spikes by returning to an earlier checkpoint and skipping batches [23]. OPT combined checkpoint recovery with learning-rate reduction and a tighter gradient-clipping threshold [24]. These interventions act on both the magnitude of the update and the state from which training continues.

Recent open-weight reports also emphasize prevention. OLMo 2 tracks spikes in loss and gradient norm while evaluating initialization, normalization, and optimizer choices [25]. Kimi K2's MuonClip optimizer uses attention-logit-based rescaling of query and key weights and reports training without loss spikes [26].

Gradient norms are thus already useful monitoring signals. Their relationship to stability becomes clearer when magnitude is combined with direction: a growing gradient can reinforce or oppose its predecessor. We develop this connection through constructive and destructive regimes, showing how consecutive gradients estimate directional curvature and reveal the conditions for a local loss increase. This provides a geometric interpretation of signals available during training, using one retained gradient and no Hessian computation.

III. Constructive and Destructive Regimes

Loss spikes and oscillatory updates motivate diagnostics that explain how training is progressing as it happens. The gradients already computed by the optimizer contain two complementary signals: how strongly the loss changes locally, and whether the next update reinforces or opposes the preceding one. We first show how these interactions determine accumulated parameter change. We then connect consecutive alignment to curvature, obtaining a secant estimator and a joint condition on gradient reversal and growth that explains when the loss can increase.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{6}$$

For training from scratch, θ_0 is the initial model and θ_T contains the parameter changes accumulated while learning from the data. Expanding their squared difference separates individual step magnitudes from interactions across steps:

$$\begin{aligned}\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t.\end{aligned} \tag{7}$$

The squared norms are nonnegative contributions from individual steps. In full-batch GD, the negative gradient is the direction of steepest local descent, and a larger norm means a steeper local slope. The cross-step terms determine how these changes accumulate. Positive inner products reinforce displacement; negative inner products produce *destructive interference*, with one update undoing a component of another. Large individual steps can therefore coexist with limited net parameter change.

To make the respective roles of magnitude and direction explicit, factor each inner product as $g_u^\top g_t = \|g_u\|\|g_t\|\rho_{u,t}$, where

$$\rho_{u,t} := \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \qquad 1 \le u < t \le T. \tag{8}$$

For nonzero consecutive gradients, write $\rho_t := \rho_{t-1,t}$. Substitution into Eq. (7) gives

$$\begin{aligned}\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} \|g_u\|\|g_t\|\rho_{u,t}.\end{aligned} \tag{9}$$

Constructive and destructive behavior are thus encoded directly in the SGD update equations. For fixed norms, positive alignment increases net displacement, whereas negative alignment limits it through cancellation. We identify a *constructive regime* by persistent $\rho_t > 0$ and a *destructive regime* by persistent $\rho_t < 0$.

A. Directional Curvature and Consecutive Alignment

We now connect these regimes to the curvature encountered along a step. For mini-batch SGD, use $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$ from Eq. (4). Suppose $\mathcal{L}$ is twice continuously differentiable along the preceding step and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} gives

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{10}$$

where r_t is the Taylor remainder. Neglecting it and projecting onto the preceding gradient yields

$$\begin{aligned}\|g_{t-1}\|\|g_t\|\rho_t \approx \quad & \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{- \eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ & \underbrace{+ g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}.\end{aligned} \tag{11}$$

Define the curvature along the preceding gradient and the normalized noise contribution as

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{12}$$

$$\varepsilon_t := \frac{g_{t-1}^\top (\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{13}$$

Dividing Eq. (11) by the two gradient norms gives the central relation

$$\boxed{\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t).} \tag{14}$$

In full-batch GD, $\varepsilon_t = 0$ and the norm ratio is positive. Within this approximation, $\rho_t \propto 1 - \eta\kappa_t$, so $\eta\kappa_t < 1$ gives constructive alignment, while $\eta\kappa_t > 1$ gives destructive alignment. The boundary $\eta\kappa_t = 1$ separates the two regimes. At this boundary the gradients are orthogonal if both are nonzero; a vanishing gradient leaves the cosine undefined.

Destructive alignment means that an update opposes a component of the preceding parameter displacement. It does not establish overwriting of learned representations, which would require a functional measurement. From an optimization perspective, negative alignment indicates that a GD step has passed the minimum along its preceding direction in the local quadratic model. The optimizer then partially reverses that step. Such reversals can coexist with loss reduction; within the same approximation, the returning gradient's magnitude determines the loss sign.

B. Estimating Curvature without Hessian Computation

Equation (14) turns consecutive training gradients into a directional-curvature estimator:

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx \eta\kappa_t - \varepsilon_t. \tag{15}$$

Its computation requires retaining one additional gradient and taking norms and an inner product. Thus curvature information can be monitored at every training step using gradients the optimizer already needs.

The measured projected change has a precise geometric meaning. In smooth full-batch GD, $\widehat{\kappa}_t$ is exactly the directional Hessian curvature averaged along the preceding step; it approximates the starting-point Rayleigh quotient κ_t when curvature varies little over that step. Appendix A gives the identity and an error bound. It measures curvature along the realized gradient direction, while $\lambda_{\max}(H_t)$ measures the sharpest direction. In SGD, the additional term $-\varepsilon_t$ incorporates sampling. Larger batches reduce the gradient sampling variance and can bring the diagnostic closer to its full-batch interpretation. Its importance still depends on gradient variability relative to the full gradient, which we examine next.

C. Noise Contribution under Mini-Batch Sampling

Fresh, conditionally unbiased batches satisfy $\mathbb{E}[g_{t-1}^\top \xi_t] = 0$ and $\mathbb{E}[\nabla\mathcal{L}(\theta_{t-2})^\top \xi_{t-1}] = 0$. Consequently, the noise contribution to consecutive alignment has nonpositive expectation:

$$\mathbb{E}[g_{t-1}^\top (\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{16}$$

For uniform batches sampled without replacement,

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)} \mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{17}$$

Smaller batches therefore increase the magnitude of this negative unnormalized contribution at fixed parameters; full-batch evaluation eliminates it.

For two independent gradients sampled at the same fixed state θ, concentration of their norms gives the approximation

$$\mathbb{E}[\rho_t \mid \theta \text{ fixed}] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)} \operatorname{tr}\Sigma(\theta)}. \tag{18}$$

Reducing b weakens this shared directional signal, moving the expected cosine toward zero. Along an evolving SGD trajectory, sampling also changes the next parameter state and its curvature. The negative term in Eq. (16) alone therefore does not determine how a negative cosine changes. Section IV-D measures the combined sampling and trajectory effects on observed alignment during training. The same-state approximation above has a nonnegative numerator and cannot test attenuation of a negative full-gradient cosine between distinct states.

D. Normalized Displacement and Cumulative Interference

How much of the distance traveled during training remains as net parameter change? We measure this fraction by the normalized displacement

$$R_T := \frac{\|\Delta_T\|}{\eta \sum_{t=1}^T \|g_t\|} = \frac{\|\sum_{t=1}^T g_t\|}{\sum_{t=1}^T \|g_t\|}, \qquad 0 \le R_T \le 1, \tag{19}$$

defined for nonzero total path length. For constant-step plain SGD, the denominator is the traveled distance and the numerator is displacement from initialization. Equation (9) shows that R_T^2 incorporates the interactions between every pair of steps. Thus R_T summarizes the cumulative effect of their magnitudes and alignment, while ρ_t identifies the local regime. It measures parameter-path geometry, not a fraction of useful learning retained.

Aligned steps give $R_T = 1$. Equal opposing pairs give zero, and equal-length orthogonal steps give $R_T = 1/\sqrt{T}$, or about 1.83% after 3,000 updates. A small R_T therefore does not by itself imply negative cumulative interaction. With the observed gradient lengths, the zero-cross-term reference and its associated energy ratio are

$$\begin{aligned} R_T^\perp &:= \frac{\sqrt{\sum_{t=1}^T \|g_t\|^2}}{\sum_{t=1}^T \|g_t\|}, \\ Q_T &:= \left(\frac{R_T}{R_T^\perp}\right)^2 = 1 + \frac{2\sum_{u<t} g_u^\top g_t}{\sum_{t=1}^T \|g_t\|^2}. \end{aligned} \tag{20}$$

For nonzero gradient energy, $Q_T < 1$, $Q_T = 1$ and $Q_T > 1$ mean negative, zero and positive total cross-step interaction, respectively. The reference is algebraic; it does not assume random independent steps.

To locate cancellation across timescales, combine adjacent updates. For $T = 2J$, let $h_j = g_{2j-1} + g_{2j}$, so that $\theta_{2j} - \theta_{2j-2} = -\eta h_j$. Define

$$C_T := \frac{\sum_{j=1}^{J} \|h_j\|}{\sum_{t=1}^{T} \|g_t\|}, \quad D_T := \frac{\|\sum_{j=1}^{J} h_j\|}{\sum_{j=1}^{J} \|h_j\|}, \qquad R_T = C_T D_T. \tag{21}$$

Here C_T measures the path length remaining after cancellation within pairs; D_T measures how coherently those pair displacements accumulate. Both lie in $[0, 1]$ when their denominators are nonzero. If the original path length is positive but all pair sums vanish, $C_T = R_T = 0$ and D_T is undefined. Each pair norm follows from the raw norms and cosine,

$$\|h_j\|^2 = \|g_{2j-1}\|^2 + \|g_{2j}\|^2 + 2\rho_{2j}\|g_{2j-1}\|\|g_{2j}\|. \tag{22}$$

The factorization distinguishes immediate cancellation from subsequent turning; neither factor alone determines whether the objective decreases.

E. Conditions for Loss Decrease and Spike Formation

When does a destructive regime produce a loss increase? Define the full-objective change over the preceding step as $\Delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. The second-order approximation derived in Appendix A separates three contributions:

$$\begin{aligned} \Delta\mathcal{L}_{t-1} \approx\ & \underbrace{-\frac{\eta}{2}\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\frac{\eta}{2}\rho_t\|g_{t-1}\|\|g_t\|}_{\text{Directional interference}} \\ & + \underbrace{\frac{\eta}{2} g_{t-1}^{\top}(\xi_{t-1} + \xi_t)}_{\text{Noise interaction}}. \end{aligned} \tag{23}$$

The first term favors descent. Constructive alignment adds a further decrease, whereas destructive alignment offsets part of it. The size of this opposing contribution depends jointly on reversal and gradient growth. In full-batch GD with nonzero gradients, the approximation predicts $\Delta\mathcal{L}_{t-1} > 0$ precisely when

$$\boxed{\rho_t < -\frac{\|g_{t-1}\|}{\|g_t\|}.} \tag{24}$$

Two conditions are necessary: $\rho_t < 0$ and $\|g_t\| > \|g_{t-1}\|$. The new gradient must reverse direction strongly enough that its opposing component exceeds the preceding magnitude. For example, doubling the norm requires $\rho_t < -0.5$, an angle above $120°$. A fivefold increase requires only $\rho_t < -0.2$, or about $101.5°$. Greater norm growth therefore makes even a milder reversal sufficient for an upward loss step.

The equivalent curvature form is

$$\Delta\mathcal{L}_{t-1} \approx \frac{\eta}{2}\|g_{t-1}\|^2(\eta\kappa_t - 2). \tag{25}$$

It summarizes the same joint effect in the quadratic model: $\eta\kappa_t = 1$ marks the alignment boundary, while $\eta\kappa_t = 2$ marks its loss-change threshold. Between them, destructive updates still decrease the quadratic objective. For the measured secant statistic, crossing one is exactly equivalent to negative consecutive cosine; crossing two predicts the loss sign only under the stated approximation. The expressions are exact for quadratic objectives; for nonlinear losses their accuracy depends on variation along the step, with sampling adding the explicit noise contribution. A sequence of upward steps followed by recovery forms the spike events studied below. The diagnostic thus connects local curvature to the reversal and norm growth underlying their formation.

IV. Gradient Reversal and Progress in GD and SGD

Full-batch GD can continue reducing the training objective while successive gradients become nearly antiparallel. In each of the three controlled learning-rate sweeps below, the largest tested rate reaches the lowest training loss after 3,000 updates, although net displacement is only 2.6–5.4% of path length. We ask how much motion survives adjacent reversals, how that motion accumulates, and how it accompanies loss reduction. We then examine the alignment observed during SGD.

Oscillatory descent at the edge of stability is established by Cohen et al. [13], with self-stabilization analyzed by Damian et al. [15]. Singh et al. [16] study directionality at the trajectory and individual-update scales. Our measurements quantify the distinction between local reversal, cumulative interaction and objective progress using the training gradients.

a) Controlled design.: We train a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 training images, and a word-level Transformer on WikiText-2. Each minimizes its fixed objective for 3,000 full-batch updates. Constant learning rates are $\{0.004, 0.02, 0.04\}$ for the MLP, $\{0.025, 0.05, 0.1\}$ for ResNet and $\{0.05, 0.1, 0.2\}$ for the Transformer. Within each model, the rates share one frozen initialization and data preparation. The comparison is conditional on that initialization; the later batch-size study uses three initialization blocks. Appendix B specifies the fixed objectives, parameter scope and numerical conventions.

A. The Onset of Sustained Gradient Reversal

Figures 1 and 2 show the transition from positively aligned updates to oscillatory loss reduction. The lowest MLP and Transformer rates give smooth loss curves, positive consecutive cosine and $\eta\widehat{\kappa}_t$ near zero. The other seven conditions develop nearly opposing consecutive gradients while the loss continues to decrease over longer timescales. ResNet exhibits reversal at every tested rate, with an earlier transition at larger rates. The highest-rate MLP also has a brief initial reversal before its later sustained negative phase.

The secant statistic expresses this geometry in curvature terms. By Eq. (15), nearly opposite gradients with similar norms necessarily give $\eta\widehat{\kappa}_t$ near two. Over the last 1,000 observations in the seven reversing conditions, mean adjacent norm ratios range from approximately 1.000 to 1.003 and mean scaled secants from 1.97 and 1.99. These are linked descriptions of the measured gradients, not independent confirmation of a mechanism. One marks the exact sign change of consecutive alignment; two is the quadratic-model loss-change

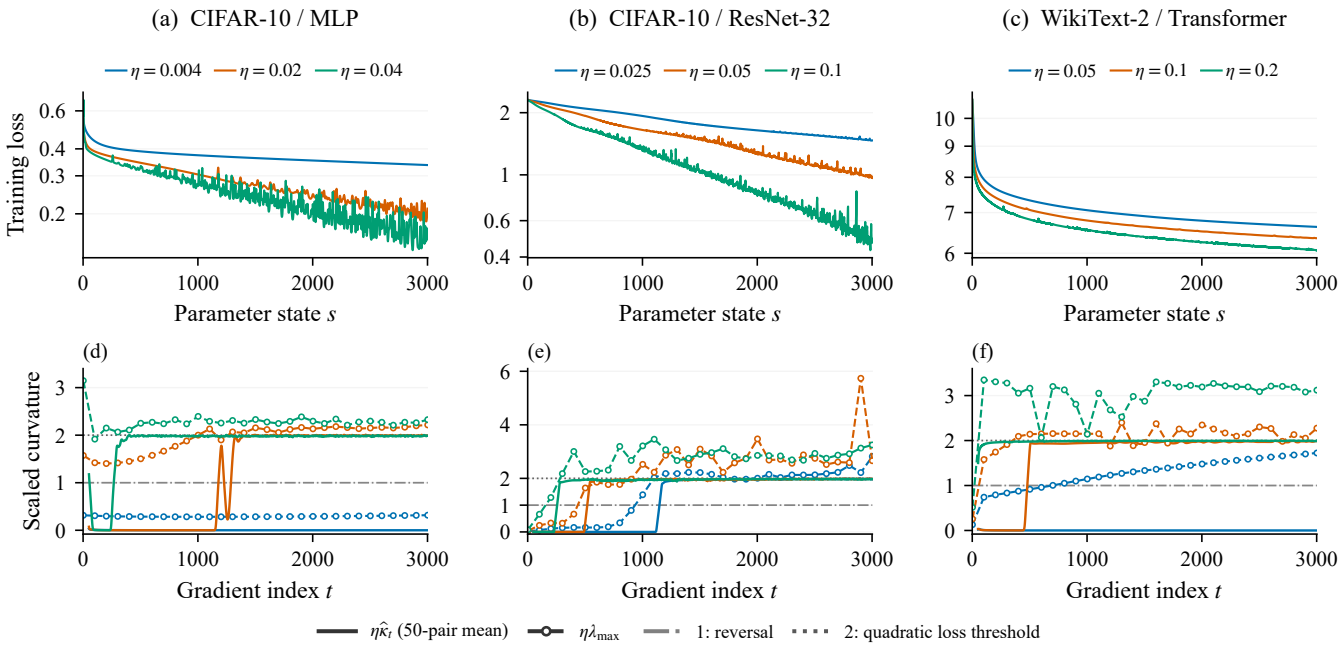


Fig. 1: **Oscillatory loss reduction and directional curvature in GD.** Columns identify dataset/model; colors identify learning rate. Top: unsmoothed full-objective loss on logarithmic axes. Bottom: $\eta\widehat{\kappa}_t$ (solid, complete 50-pair trailing means) and recorded $\eta\lambda_{\max}$ (dashed with circles). Reference lines mark the alignment threshold at one and the quadratic-model loss-change threshold at two; their stepwise interpretation applies before smoothing. Spectra use the complete MLP objective, 5,000/50,000 ResNet images and 2,500/59,680 Transformer sequences. They are pointwise maxima; the secant is a directional path average under the smooth full-GD interpretation. Loss uses state s; a spectrum at state s is placed at $t = s + 2$, the starting state for $\widehat{\kappa}_t$. The final spectrum has no recorded gradient pair.

threshold. The displayed 50-pair trailing means summarize regimes, whereas those thresholds apply to the unsmoothed stepwise quantities.

The Hessian checkpoints provide spectral context. They measure maximal rather than gradient-direction curvature, at a parameter state rather than averaged along an update under the smooth full-GD interpretation. For ResNet and Transformer they also use fixed subset objectives. These three differences prevent attributing the numerical gap solely to gradient orientation or using it as an independent estimator-fidelity test. The secant statistic remains available at every update without additional gradient evaluations or Hessian-vector products.

B. Local Reversal Coexists with Substantial Loss Reduction

In Fig. 2, reversal accompanies growing gradient norms for the MLP and ResNet. These larger opposing steps accumulate path length faster than displacement from initialization. At the highest tested rates, R_{3000} is 2.6%, 3.7% and 5.4% for the MLP, ResNet and Transformer, respectively. Yet their final training losses, 0.1539, 0.4898 and 6.0697, are the lowest within their respective sweeps. Persistent local reversal therefore coexists with substantial training-loss reduction; its sign alone does not establish a causal benefit or an asymptotic convergence result.

The zero-cross-term reference makes these small path ratios interpretable. At the highest rates, $R^{\perp}_{3000}$ is 1.99%, 2.02% and 1.84%, respectively. The measured ratios exceed these references, giving $Q_{3000} = 1.66$, 3.43 and 8.67. Indeed, $Q_{3000} > 1$ in all nine conditions: total interaction across all pairs is positive even though cumulative adjacent interaction is negative in the seven reversing runs. Here "destructive" identifies opposing local displacements; a small R_T measures path geometry, not lost learning or wasted computation.

Pairing the updates locates this cancellation more precisely. At the highest rates, C_{3000} is 0.114, 0.156 and 0.086, while D_{3000} is 0.225, 0.240 and 0.630. Thus adjacent pairing removes most of the original path length. The smaller residual displacements accumulate more coherently in the Transformer than in the other two cases, where substantial further turning remains. Shifting the pairing by one update, with boundary singletons retained, changes each factor for these three runs by less than 0.004 in absolute value (Table II).

This geometric accounting does not imply descent after every pair. Over states 2000–3000, the full-objective loss decreases by 0.0644, 0.3048 and 0.1867, respectively, yet only 49.2%, 49.6% and 77.2% of the 500 nonoverlapping two-update changes are negative. The alternate offset gives 50.9%, 48.9% and 74.9% over 499 interior pairs, with a negative total change in each case. Figure 10 retains both offsets' unsmoothed changes. The residual motion accompanies net objective reduction across the window without guaranteeing the sign of each local loss change.

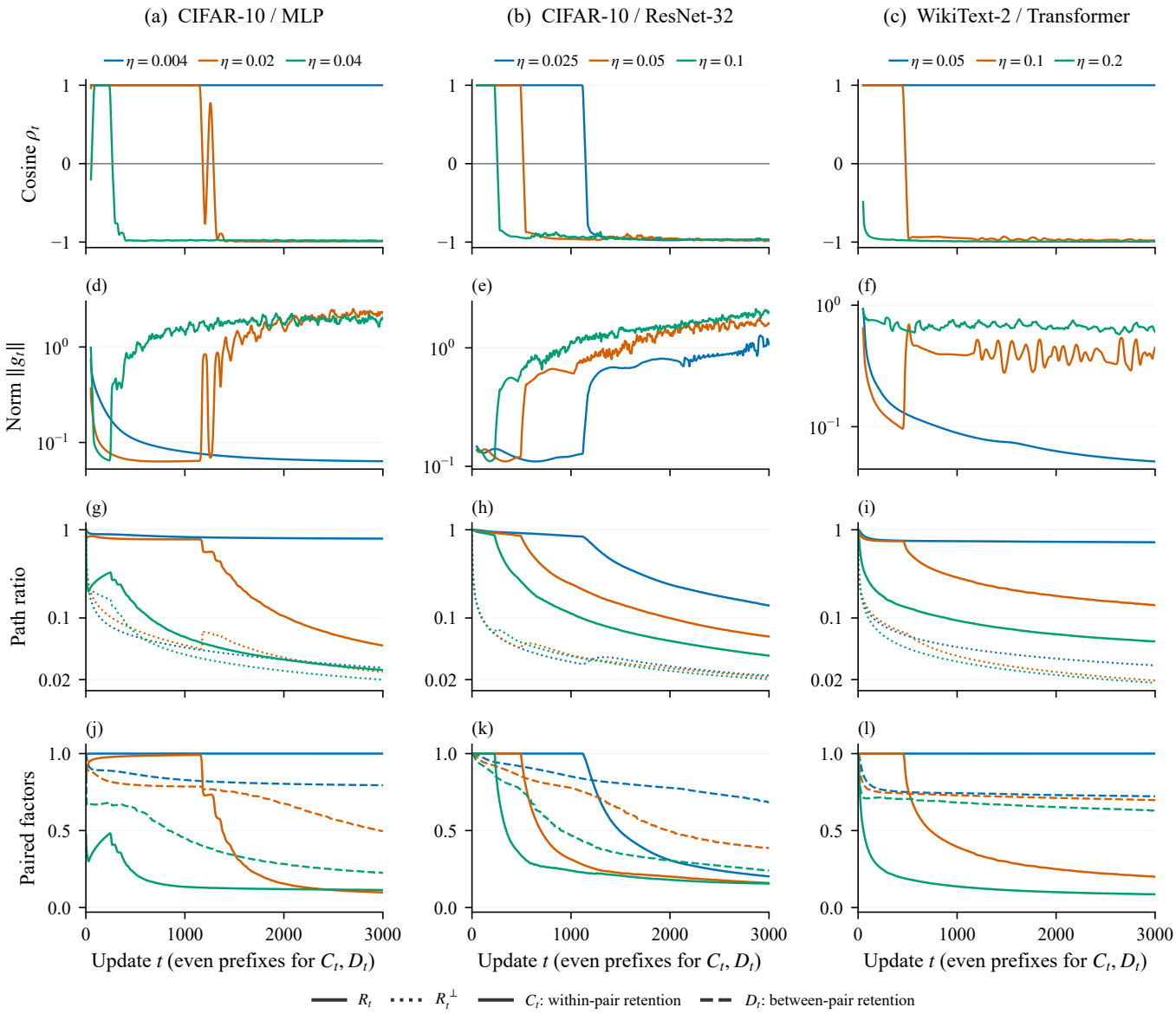


Fig. 2: **Local reversal, cumulative displacement and paired motion.** Rows show consecutive cosine, gradient norm (log scale), R_t with its zero-cross-term reference $R_t^{\perp}$ (log scale), and the factors C_t and D_t in Eq. (21). Cosines and norms use complete 50-observation trailing means; ratios use raw prefix sums, with paired factors at every even prefix. Colors denote the same rates as Fig. 1. At the largest rates, R_{3000} is 2.6%, 3.7% and 5.4% for MLP, ResNet and Transformer, yet each attains the lowest final training loss within its sweep. All nine endpoint ratios exceed their zero-cross-term references. Adjacent pairing removes most path length in the reversing runs; the remaining pair displacements undergo further turning, summarized by D_t. One initialization per model; no uncertainty band or generalization claim.

C. Reversal Has Structure across Temporal Lags

Figure 3 fixes the gradient after 2,256 updates and compares it with the preceding 256 gradients, using the largest rate for each model. The shortest lags alternate between negative odd-lag and positive even-lag alignment. The envelope contracts more rapidly for the MLP and ResNet than for the Transformer, and can strengthen again at longer lags.

The same analysis at endpoint states 756 and 3256 gives negative odd-lag and positive even-lag means over lags 1–32 in all nine model–endpoint combinations. For the Transformer, every one of these 32 lags has the corresponding sign at all three endpoints. The MLP and ResNet have sign exceptions within that range as their envelopes contract or change phase. These are three sampled endpoints in separately recomputed continuations from saved states, not exhaustive temporal coverage or exact replays of the earlier trajectories.

Lag correlations establish recurring organization. Alternation alone cannot establish net motion: the sequence $v, -v, v, -v, \ldots$ has the same ideal odd–even pattern but zero displacement after each pair. The paired factors above supply

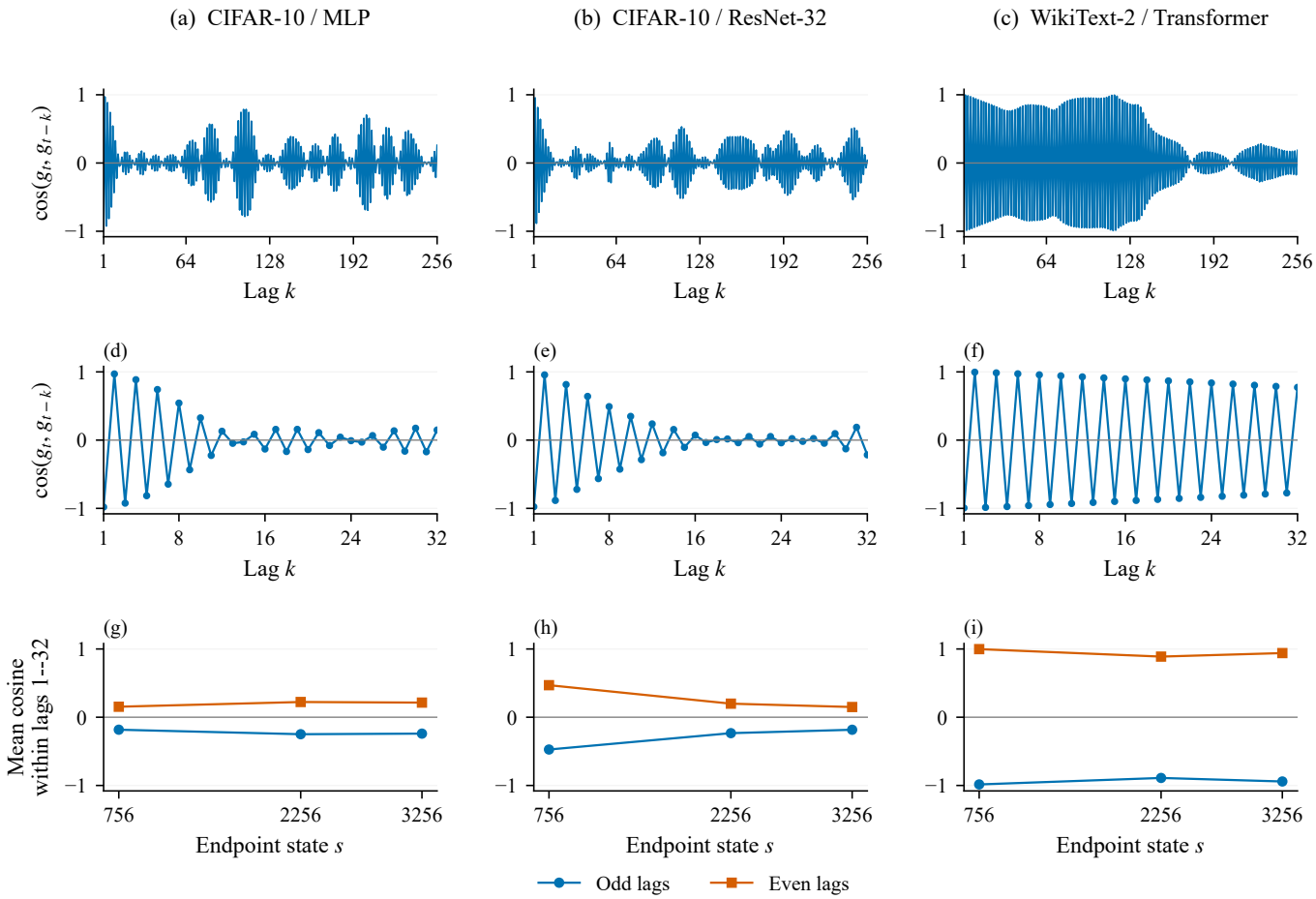


Fig. 3: **Recurring reversal at three sampled endpoints.** Top: exact backward cosine $\cos(g_t, g_{t-k})$ at fixed endpoint θ_{2256} ($t = 2257$), for lags 1–256. Middle: the same values for lags 1–32. Bottom: separate odd/even means within lags 1–32 at states 756, 2256 and 3256; connecting lines guide the eye. Rates are 0.04 (MLP), 0.1 (ResNet) and 0.2 (Transformer). Each profile comes from a separate 256-update GD continuation from saved state 500, 2000 or 3000, with one initialization. Exact products use complete raw gradients with FP64 reductions; no smoothing or endpoint averaging is applied. Later floating-point paths may differ from the original GD runs. Alternating correlations describe temporal structure; net movement also depends on gradient magnitudes.

the missing magnitude accounting, while loss measurements establish objective reduction separately.

D. Observed Alignment under Mini-Batch Sampling

Figure 4 compares 200-update continuations from initialization and from states reached after 2,000 full-GD updates. Both the reference trajectories and their continuations use constant learning rates 0.02, 0.1 and 0.2 for the MLP, ResNet and Transformer, respectively. Each stochastic condition has eight sampling repetitions in each of three initialization blocks; full GD has one continuation per block. We average the 199 adjacent cosines within a continuation, then repetitions within a block, then the three blocks equally. Shading shows the block range, not a confidence interval.

From the trained states, full-GD mean cosines are -0.991, -0.967 and -0.993. The smallest tested batches move them toward zero, to -0.384, -0.612 and -0.258, respectively. This ordering holds in every initialization block. From initialization, full GD is positively aligned for the MLP and ResNet (0.994 and 0.999), whereas the Transformer already develops reversal within the 200-update window, giving a mean of -0.827. The smallest ResNet batch reduces the positive mean to 0.418; the smallest Transformer batch gives -0.259. MLP batches of 32–512 instead change the mean sign to negative in all three blocks, with a nonmonotone response across these smaller batches. Batch size therefore changes the observed signature differently across starting states and models.

Each update draws a fresh uniform subset without replacement, independently across updates. Within a repetition, batch sizes share prefixes of the same random permutation. MLP batches count images; ResNet samples whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Thus $N = 50$ and b is the number of groups for ResNet in Eq. (17), although the figure labels image counts. Appendix B specifies the Transformer token weighting and runtime qualification. Equal update budgets imply unequal

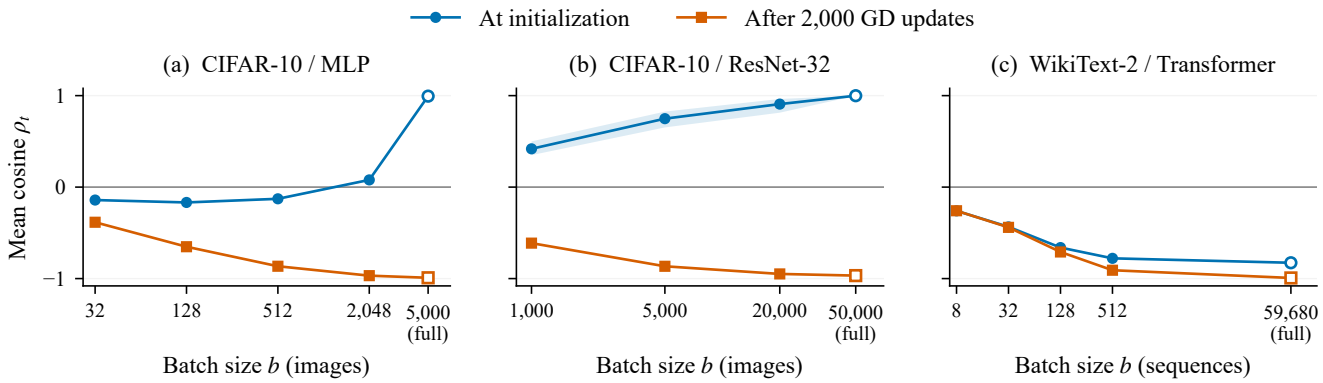


Fig. 4: **Batch-size dependence of observed alignment during SGD.** Mean sampled-gradient cosine over 199 adjacent pairs in 200 updates. Blue circles start at initialization; orange squares start after 2,000 GD updates. The reference trajectories and continuations use rates 0.02, 0.1 and 0.2 for MLP, ResNet and Transformer. Each line averages eight stochastic repetitions within initialization, then weights three initialization blocks equally; shading is their range. Hollow rightmost markers denote one full-GD control per block. ResNet batches contain whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Batch axes are logarithmic. Parameters evolve during each continuation, so the comparison includes both sampling variability and changes in the visited states.

data exposure.

These are moving trajectories. Changing batch size affects both the sampled gradients and the subsequent parameter states, so weaker sampled cosine does not establish weaker underlying full-gradient reversal. A control that independently samples at two frozen endpoints with known full-gradient alignment would isolate measurement attenuation; the present experiment measures the combined sampling and trajectory effects. The same-state approximation in Eq. (18) cannot supply a negative-alignment reference for that control.

Observed alignment also does not order objective progress. For the Transformer at $b = 8$, mean cosine is about -0.26 from both starts, but the mean full-training-loss change is -3.505 from initialization and $+0.064$ from the trained state. Together with the GD results, this separates a local directional signature from cumulative motion and objective reduction.

V. Raw-Gradient Diagnostics under Adam

Adam [27] transforms the raw gradient through first- and second-moment estimates,

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{26}$$

with zero initial moments. Bias correction gives $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$, followed by

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad \theta_t = \theta_{t-1} - \eta d_t, \tag{27}$$

where operations are coordinatewise. Throughout this section, g_t is the raw loss gradient and ρ_t remains the cosine between consecutive raw gradients, as in Section III. We evaluate $\eta\widehat{\kappa}_t$ using the same diagnostic in Eq. (15). Although d_t determines the step, g_t describes the local loss slope. Changes in its direction and norm reveal how that slope varies across the states visited by Adam. Comparing raw-gradient and first-moment measurements then shows how this response propagates through the optimizer's memory. Momentum cosines are labeled separately in the figures.

We reuse the CIFAR-10/MLP, CIFAR-10/ResNet-32 and WikiText-2/Transformer objectives from Section IV. Each Adam trajectory starts from the same frozen, untrained seed-0 weights used for its GD comparison, with fresh moments. We run 1,024 full-objective updates at each constant rate: $\{0.0001, 0.0003, 0.001\}$ for the MLP and $\{0.0003, 0.001, 0.003\}$ for ResNet and Transformer. All nine arms use $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$, without a schedule, clipping or weight decay. There is one initialization per model, matched across rates. Full-batch accumulation removes sampling variation in the objective; model, objective and numerical details are in Appendix B-D.

Figure 5 shows overall convergence with transient loss spikes, especially at larger rates. The largest tested rate attains the lowest final training loss in each model despite these excursions. These excursions interrupt trajectories whose longer-term loss continues to decrease. We next examine their gradient dynamics before summarizing the spike distribution across the nine arms.

A. Anatomy of Individual Spikes

For each model, we select the recovered excursion with the largest peak/pre-rise loss ratio at its highest tested rate. Selection uses only loss, with no condition on a diagnostic. The selected peaks occur at states 703, 422 and 794 for MLP, ResNet and Transformer, respectively.

All three examples in Fig. 6 enter the pronounced rise with negative raw-gradient cosine. During the rise, the gradient norm increases sharply and $\eta\widehat{\kappa}$ exceeds two before the loss maximum. Momentum norms also grow.

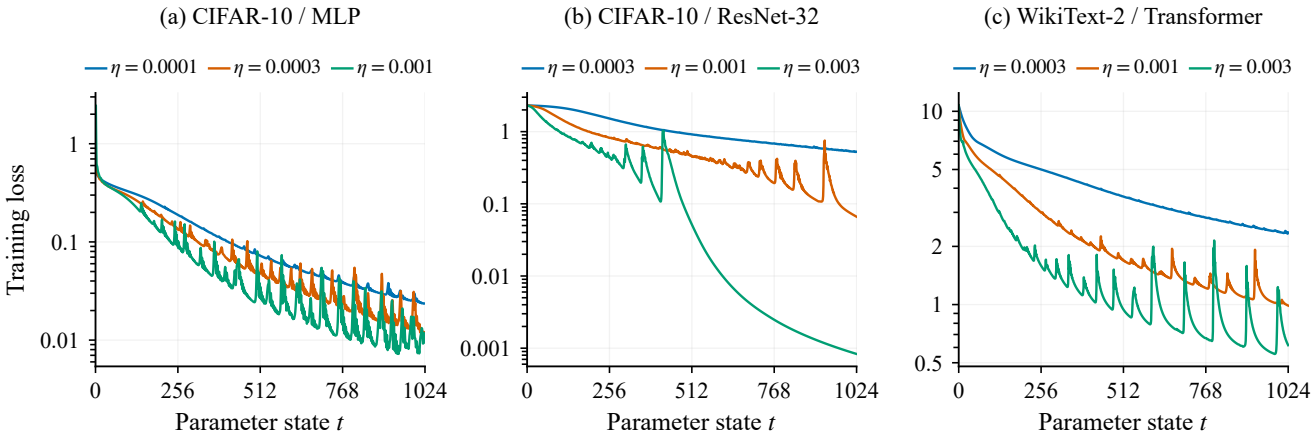


Fig. 5: Full-batch Adam training from matched initial weights, with three constant learning rates per model. Every training-loss observation is shown without smoothing on logarithmic y-axes. Parameter state s denotes s completed updates, from initialization to 1,024 updates. One initialization per model; colors identify increasing rates within each panel.

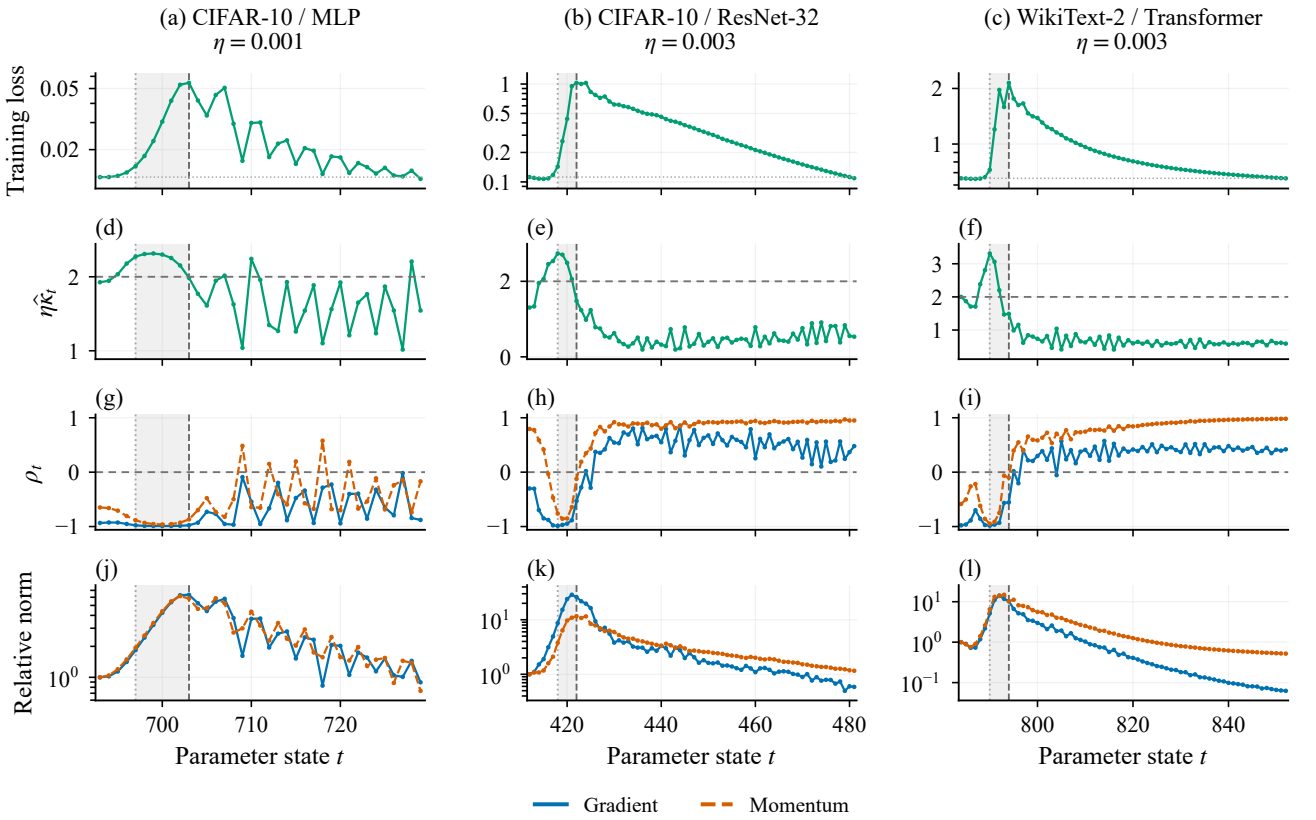


Fig. 6: Largest recovered relative loss spike at each model's highest Adam rate. Rows show loss, raw-gradient projected change, consecutive cosines of raw gradients and first moments, and their relative norms. All observations are unsmoothed. At parameter state s, vector diagnostics refer to g_{s+1} and m_{s+1}. Norms are divided by their value at the first displayed state. Gray bands mark the pronounced rise from the first +10% crossing to the loss peak (697–703, 418–422 and 790–794), rather than the entire excursion through recovery. The horizontal dotted loss reference is the first displayed loss. Windows begin ten states before the peak and end at the first later loss below that displayed reference.

The descent separates the models. ResNet and Transformer return to constructive regimes, with positive gradient and momentum cosines, falling $\eta\widehat{\kappa}$ and a comparatively smooth loss decline. The MLP continues to oscillate: raw cosine remains mostly negative, momentum cosine alternates between signs, and loss and norms decrease through repeated rebounds. Together, the examples expose developing spikes through destructive alignment and sharp gradient growth. The early increase in $\eta\widehat{\kappa}$ motivates intervention before the loss reaches its maximum.

B. Loss-Defined Events and Their Distribution

To examine repetition without selecting on gradients, write $\mathcal{L}_s := \mathcal{L}(\theta_s)$ for the recorded training loss. Starting at initialization, let a_i be the earliest available state with $\mathcal{L}_{a_i+1} > \mathcal{L}_{a_i}$. Hold the pre-rise reference $\mathcal{L}_{a_i}$ fixed. Recovery occurs at the first later state whose loss is at or below this pre-rise level, $\mathcal{L}_s \le \mathcal{L}_{a_i}$. For an event recovered within T updates, define

$$\begin{aligned} c_i &:= \min\{s : a_i < s \le T,\ \mathcal{L}_s \le \mathcal{L}_{a_i}\}, \\ p_i &\in \operatorname*{arg\,max}_{a_i<s<c_i} \mathcal{L}_s. \end{aligned} \tag{28}$$

If several states attain the maximum, p_i is the earliest one. Figure 7 illustrates these boundaries. An event is included only when recovery is observed by $T = 1024$. Search resumes at c_i; partial declines and additional local maxima before this return belong to the same event. Thus a_i is the last pre-rise state, $a_i + 1$ the first elevated state, p_i the loss peak, and c_i the first recovered state. Peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$; rise and recovery times are $p_i - a_i$ and $c_i - p_i$, in optimizer updates. The three loss ratios are 1, $\mathcal{L}_{p_i}/\mathcal{L}_{a_i}$ and $\mathcal{L}_{c_i}/\mathcal{L}_{a_i} \le 1$.

Figure 8 places the illustrations within the complete distribution of recovered excursions. Small fluctuations coexist with long tails; larger rates populate more severe excursions, although the distributions overlap and do not undergo a uniform shift. To focus the anatomy analysis on pronounced changes, we retain events with peak increase strictly greater than 10%. The filtered sample contains 129 recovered events. Two further above-threshold candidates without observed recovery are excluded.

For Table I, pre-rise diagnostics are measured at θ_{a_i}, hence use gradient index $a_i + 1$. Recovery entries first average each event's diagnostic over the arrival states $p_i + 1, \dots, c_i$, then summarize those means across events with equal event weight. They are not values only at c_i. The norm-ratio columns instead use the endpoints $\|g_{p_i+1}\|/\|g_{a_i+1}\|$ and $\|g_{c_i+1}\|/\|g_{p_i+1}\|$. The gradient norm's own maximum need not coincide with the loss peak.

Among qualifying events, both the mean increase and the maximum increase grow with LR in every model with more than one eligible rate. The largest increase is 858.8% for ResNet at $\eta = 0.003$. Mean recovery takes 17–19 updates versus 6–8 for the ResNet rise, and 24–31 versus about 6 for the Transformer. MLP rise and recovery times are more comparable. Thus, a short amplification can be followed by a much longer descent, without implying that all intermediate loss changes are monotonic.

Pre-rise raw cosine is negative in 92.1% of events with a defined pre-rise measurement. The gradient norm at the loss peak exceeds its pre-rise value in 100% of events. Projected change is already above two before the first loss increase in 16.5% of defined cases. Its full-event maximum precedes the loss peak in 63.6% of events, coincides in 22.5% and follows in 14.0%.

C. Reversal, Norm Amplification and Possible Interventions

To relate the initial regime to amplification during an event, we plot

$$\rho_{a_i+1}\frac{\|g_{p_i+1}\|}{\|g_{a_i+1}\|} \quad \text{against} \quad \frac{\mathcal{L}_{p_i}}{\mathcal{L}_{a_i}}. \tag{29}$$

This combines the pre-rise consecutive cosine with the subsequent rise norm ratio. Figure 9 shows that a destructive initial regime combined with gradient amplification explains a substantial part of spike magnitude: the per-rate fits account for 67–69% of log-loss-ratio variation for the MLP, 90–94% for ResNet, and 38–54% for the Transformer. This relationship connects the individual spike examples to the broader event sample. Reversal identifies the regime; gradient growth explains how strongly that reversal is amplified.

This anatomy motivates testing whether intervention during reversal and early norm growth can attenuate the subsequent excursion. Gradient-norm clipping was proposed to address exploding gradients [28], and analyses allowing smoothness to grow with gradient norm provide a further rationale for adaptive scaling [29]. Learning-rate adjustment and data-mixing interventions provide other tests of the proposed interaction between the current regime and subsequent gradient growth. These tests should isolate states before pronounced excursions, compare matched interventions with unchanged continuations, and measure both spike attenuation and later loss at equal budgets. Establishing prospective warning information also requires non-spike comparisons. Such interventions and predictive validation remain future work.

VI. Conclusion

The displacement formulation distinguishes the magnitude of individual updates, their local directional interaction and their cumulative accumulation. For plain SGD, the exact decomposition retains all temporal lags, while consecutive cosine provides a tractable local observation. The curvature and sampling relations clarify how that observation can change without determining either cumulative displacement or loss improvement. The full-batch experiments show negative consecutive interaction alongside positive total interaction across all pairs. Pairing adjacent updates identifies substantial immediate cancellation, followed by model-dependent accumulation of the residual motion. Neither factor guarantees a local loss decrease. The mini-batch continuations further show that observed alignment changes with both sampling and the evolving trajectory; weaker reversal alone does not order the attained loss.

Under Adam, raw-gradient diagnostics remain informative about the loss landscape traversed by adaptive steps. Controlled full-batch experiments connect loss spikes with a

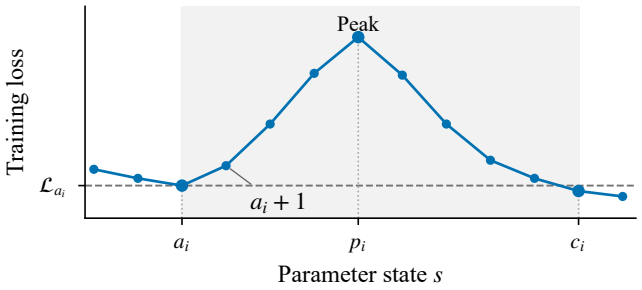


Fig. 7: Spike boundaries (schematic). The pre-rise state a_i fixes the reference loss; $a_i + 1$ is the first elevated state, p_i the peak, and c_i the first return to or below that reference. Rise and recovery durations are $p_i - a_i$ and $c_i - p_i$.

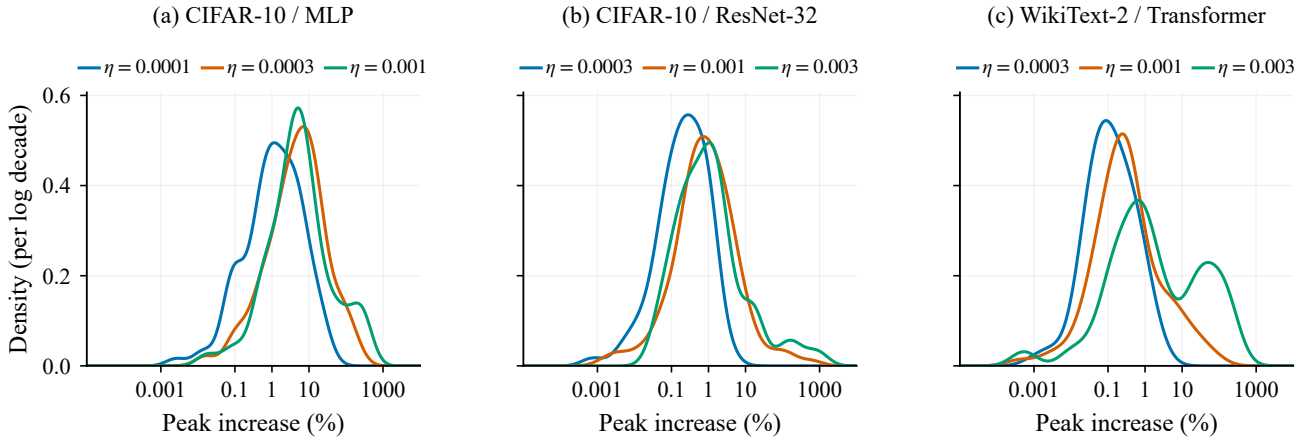


Fig. 8: Distributions of peak percentage increases for all 806 recovered excursions, including small fluctuations, with no amplitude cutoff. Colors identify LRs within each model. Gaussian kernel densities are estimated in $\log_{10}$ percentage-increase coordinates with one pooled bandwidth per model, common across its LRs. Each LR density integrates to one in that coordinate. The log x-axis shows percentage increase: 1 denotes +1%, and 100 denotes +100%. This differs from the loss-ratio axis in Fig. 9.

TABLE I: Anatomy of recovered spikes with peak increase greater than 10%. Cells report mean ± sample SD across events; maximum peak increase has its own subcolumn. Before denotes a single diagnostic at state a_i; recovery denotes the event's mean over states $p_i + 1, \ldots, c_i$. Norm ratios compare loss-peak/pre-rise and recovered/loss-peak endpoints. N counts included events; dashes indicate no qualifying event. Two initialization events have undefined pre-rise diagnostics, leaving 46 and 38 observations for those entries in the MLP's middle and highest LR rows. No smoothing or averaging across initializations.

Dataset Model	η	N	Peak increase (%)		Time (updates)		$\eta\widehat{\kappa}$ (raw)		ρ (raw)		Gradient norm ratio	
			Mean ± SD	Max	Rise	Recovery	Before	Recovery	Before	Recovery	Rise	Recovery
CIFAR-10 MLP	10^{-4}	8	19 ± 6.1	31	8.4 ± 2.9	4.4 ± 2.4	1.8 ± 0.38	1.7 ± 0.08	-0.77 ± 0.31	-0.89 ± 0.08	5.3 ± 1.7	0.30 ± 0.07
	3×10^{-4}	47	43 ± 46	203	5.5 ± 4.4	4.9 ± 4.0	1.3 ± 0.25	1.6 ± 0.18	-0.53 ± 0.29	-0.71 ± 0.24	4.5 ± 3.0	0.33 ± 0.13
	10^{-3}	39	95 ± 107	318	6.1 ± 5.9	7.4 ± 7.5	1.4 ± 0.40	1.4 ± 0.26	-0.46 ± 0.38	-0.55 ± 0.30	5.2 ± 3.8	0.31 ± 0.14
CIFAR-10 ResNet-32	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	10	105 ± 182	602	7.5 ± 2.2	17 ± 11	1.3 ± 0.39	1.3 ± 0.25	-0.36 ± 0.42	-0.36 ± 0.26	4.3 ± 2.9	0.43 ± 0.24
	3×10^{-3}	7	179 ± 309	859	6.3 ± 1.3	19 ± 20	1.7 ± 0.38	1.1 ± 0.37	-0.65 ± 0.28	-0.13 ± 0.42	5.0 ± 4.2	0.34 ± 0.24
WikiText-2 Transformer	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	6	33 ± 25	81	6.3 ± 1.9	24 ± 9.8	2.1 ± 0.27	0.96 ± 0.16	-0.84 ± 0.10	-0.02 ± 0.12	9.0 ± 3.8	0.09 ± 0.05
	3×10^{-3}	12	82 ± 70	231	6.2 ± 2.7	31 ± 17	2.2 ± 0.43	0.91 ± 0.19	-0.79 ± 0.22	0.09 ± 0.21	11 ± 6.9	0.08 ± 0.08

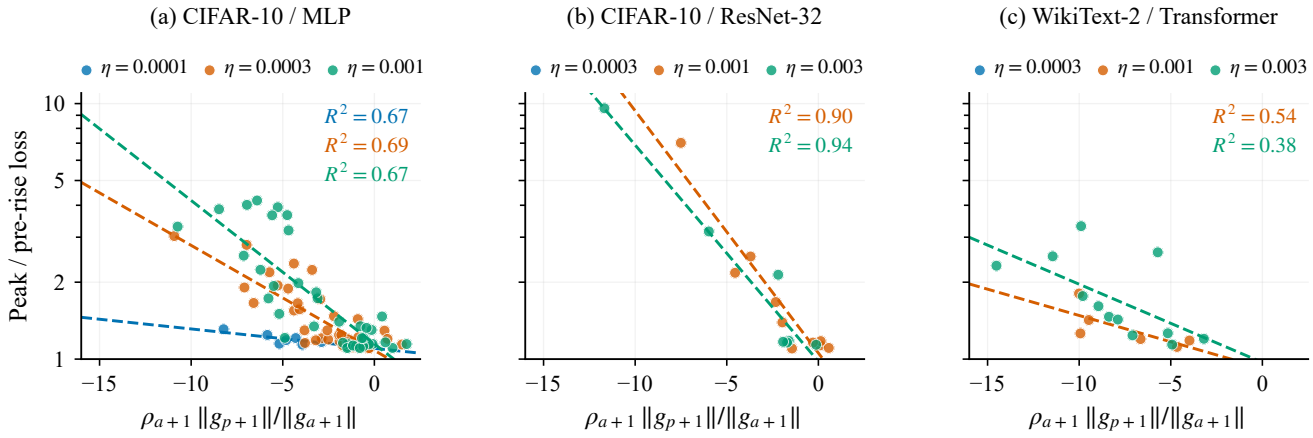


Fig. 9: Pre-rise raw-gradient cosine times rise norm ratio versus peak/pre-rise loss for 127 recovered spikes above 10%; two initialization events have undefined pre-rise cosine. One unsmoothed point per event. Colors identify LRs. Separate per-LR least-squares fits with intercept use $\log_{10}$(peak/pre-rise loss) as the response; color-matched R^2 values are descriptive in-sample scores on that log response. Dashed lines extend across each panel, including extrapolation beyond observed samples. The y-axis is a loss ratio: 1 denotes no increase and 2 denotes +100%.

predominantly destructive pre-rise gradient regime and sharp norm amplification. The micro-plots resolve projected gradient changes before the loss maximum; the event population quantifies the variation in timing and recovery across models and rates. A joint descriptor of pre-rise reversal and rise norm growth is associated with relative peak magnitude, without establishing a causal feedback mechanism or a universal instability threshold.

The practical contribution is a framework for interpreting these measurements together and using them to formulate controlled interventions. Developing a scheduler or clipping rule requires demonstrating that acting on early diagnostics improves subsequent loss at matched budgets, beyond decisions based on recent loss and known training settings. Prospective validation, controlled interventions and replication across independent runs remain future work.

Appendix A
Curvature Estimation and Local Loss Change

Write $H(\theta) = \nabla^2\mathcal{L}(\theta)$ and assume the objective is twice continuously differentiable along the plain-SGD step. Integrate its gradient derivative on $\theta_{t-2} - s\eta g_{t-1}$, add the endpoint-noise difference, and project onto $-g_{t-1}/(\eta\|g_{t-1}\|^2)$. The result is

$$\widehat{\kappa}_t = \int_0^1 \frac{g_{t-1}^\top H(\theta_{t-2} - s\eta g_{t-1}) g_{t-1}}{\|g_{t-1}\|^2}\, ds - \frac{\varepsilon_t}{\eta}. \tag{30}$$

If H is L_H-Lipschitz in operator norm on the step, its full-batch specialization satisfies

$$|\widehat{\kappa}_t - \kappa_t| \le \int_0^1 L_H s\eta\|g_{t-1}\|\, ds = \frac{L_H\eta}{2}\|g_{t-1}\|. \tag{31}$$

This bounds the difference between the measured path average and the starting-point directional curvature.

For full-batch Adam, the displacement is $-\eta d_{t-1}$. Integrating the gradient along that step gives the corresponding identity for the same raw-gradient diagnostic in Eq. (15):

$$\widehat{\kappa}_t = \int_0^1 \frac{g_{t-1}^\top H(\theta_{t-2} - s\eta d_{t-1}) d_{t-1}}{\|g_{t-1}\|^2}\, ds. \tag{32}$$

Here d_{t-1} determines the displacement, while g_{t-1} determines the projection of the gradient response. This separates the optimizer's chosen direction from the raw loss slope measured by the diagnostic.

Consider $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$ and write $\Delta\mathcal{L}_{t-1} = \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. Integrating the full gradient along the step gives

$$\Delta\mathcal{L}_{t-1} = -\eta \int_0^1 g_{t-1}^\top \nabla\mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds. \tag{33}$$

The trapezoidal approximation replaces the integrand by its endpoint average. Substituting $\nabla\mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $\nabla\mathcal{L}(\theta_{t-1}) = g_t - \xi_t$ yields

$$\begin{aligned}\Delta\mathcal{L}_{t-1} \approx -\frac{\eta}{2}\left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right) \\ + \frac{\eta}{2} g_{t-1}^\top(\xi_{t-1} + \xi_t).\end{aligned} \tag{34}$$

The substitution $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ recovers Eq. (23). The expression is exact on quadratic objectives. A locally Lipschitz Hessian bounds its error by a constant times $\|\eta g_{t-1}\|^3$.

Using $\rho_t\|g_t\|/\|g_{t-1}\| \approx 1 - \eta\kappa_t + \varepsilon_t$ and Eq. (13) gives the equivalent curvature form,

$$\begin{aligned}\Delta\mathcal{L}_{t-1} \approx -\eta\|g_{t-1}\|^2\left(1 - \frac{\eta\kappa_t}{2}\right) \\ + \eta\xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{35}$$

It also follows directly from a second-order Taylor expansion at θ_{t-2}. With zero noise, the conditions for a loss-increasing step reduce to Eqs. (24) and (25).

The noise term has a simple conditional interpretation. Given θ_{t-2}, unbiased sampling gives $\mathbb{E}[\|g_{t-1}\|^2] = \|\nabla\mathcal{L}(\theta_{t-2})\|^2 + \mathbb{E}[\|\xi_{t-1}\|^2]$ and $\mathbb{E}[\xi_{t-1}^\top g_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]$. Their first-order variance terms cancel, leaving

$$\mathbb{E}[\Delta\mathcal{L}_{t-1} \mid \theta_{t-2}] \approx -\eta\|\nabla\mathcal{L}(\theta_{t-2})\|^2 + \frac{\eta^2}{2}\mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}]. \tag{36}$$

Thus sampling contributes through curvature at second order. A nonlinear objective also contributes the finite-step remainder.

Appendix B
Essential Measurement Conventions

A. Fixed Objectives and Numerical Scope

The tanh MLP has two 200-unit hidden layers and minimizes half-squared error summed over ten outputs and averaged over the first 5,000 CIFAR-10 images. ResNet-32 minimizes mean cross-entropy over all 50,000 training images, using 50 fixed groups of 1,000 for batch normalization. Standardization uses the complete CIFAR-10 training split: coordinate-wise for the MLP and channel-wise for ResNet. Full-batch accumulation evaluates the same fixed objective at every update; augmentation and dropout are disabled.

The word-level Transformer has two layers, two heads and embedding/feed-forward dimensions of 200. Twenty corpus streams form causal chunks of at most 35 tokens, yielding 59,680 sequences and 2,088,600 supervised targets. The objective averages negative log-likelihood over valid targets, excluding padding. Architecture and the 33,278-word first-occurrence dictionary over training, validation and test text follow PyTorch example revision `13acec6d7c78` [30].

Diagnostics cover all trainable parameters. Parameters, gradients and Adam moments use FP32; reductions use FP64, with TF32 disabled. Initial unpaired cosines and zero-denominator ratios are undefined. The GD plots' trailing means summarize sustained regimes and can delay a displayed transition; Adam losses and micro-plots are unsmoothed. All claims concern training objectives. Matched tensors give one initialization per full-batch model; independent hardware executions can develop finite-precision drift.

The GD spectral comparison reuses largest algebraic Hessian eigenvalues at states $0, 100, \ldots, 3000$, with relative eigenpair residual at most 0.005. MLP spectra cover its complete 5,000-image objective; ResNet uses its first five fixed normalization groups, and the Transformer its first 2,500 sequences. These latter spectra characterize subset objectives. The earlier MLP solver requested two eigenvalues and the other runs one; the comparison consistently uses the largest.

B. Sampling and Trajectory Comparisons

Stochastic batches are independent uniform subsets without replacement; within a repetition, batch sizes share prefixes of a common permutation. ResNet samples complete normalization groups. Transformer sampling uses $N = 59{,}680$ sequences and $M = 2{,}088{,}600$ total targets: for sequence summed losses ℓ_i, the estimator is $N(bM)^{-1}\sum_{i\in B}\ell_i$. Its fixed denominator preserves the full token-weighted objective. The seed-2 smallest two ResNet batches use a different runtime, with 200-update full-GD replay parameter relative error 2.36×10^{-7}; batch and runtime remain confounded in this block. The backward-lag curves also use separate continuations, whose later floating-point paths can differ from the original GD trajectories.

C. Pairing, Boundary Treatment and Loss Changes

The primary partition uses $(g_1, g_2), \ldots, (g_{2999}, g_{3000})$. The alternate partition pairs $(g_2, g_3), \ldots, (g_{2998}, g_{2999})$ and retains g_1 and g_{3000} as singleton segments. Thus both partitions cover the same full path and have the same total displacement. Primed factors in Table II use the alternate segment-length sum in Eq. (21). No displacement norm for a cropped window is inferred by subtracting prefix norms. Pair norms use unsmoothed statistics, evaluated as $(a-b)^2+2ab(1+\rho)$ before the square root to reduce cancellation error when ρ is close to -1.

The algebra refers to $-\eta\sum_t g_t$; actual FP32 parameter updates include rounding. At update 3000 the recorded displacement norms agree with $\eta\|\sum_t g_t\|$ to relative error below 9.6×10^{-8} across all nine runs. The table reports deterministic reductions of one initialization, without a sampling uncertainty estimate.

For loss changes, the primary pairs telescope exactly over states 2000–3000. Alternate interior pairs cover states 2001–2999; their 499 changes exclude the two boundary updates. Figure 10 shows every change at the largest rate per model. Counts in the main text use strict signs. Discarding changes no larger than 10^{-6} times the larger endpoint loss leaves all primary late-window counts unchanged. It removes two near-zero alternate-offset Transformer changes; the other late-window counts are unchanged. These dependent observations describe the selected trajectories and are not independent replication units.

D. Adam Event Summaries

At state θ_s, the raw gradient and updated first moment are g_{s+1} and m_{s+1}, before parameter update $s+1$. The terminal raw gradient is evaluated without another update. Hardware varies across ResNet and Transformer learning-rate arms; initial-gradient agreement is checked, while subsequent finite-precision drift remains possible. Rate selection uses training calibration.

Two excursions above 10% without observed recovery are excluded. Two retained initialization events have undefined pre-rise diagnostics. Each event has equal weight; recovery diagnostics first average within the event, and sample SD uses denominator $n-1$. The scatter uses per-rate ordinary least squares with an intercept on log relative peak loss. Its R^2 describes the fitting sample; dashed extensions extrapolate. For the density, each model's pooled log-percentage SD times pooled $n^{-1/5}$ sets the common Gaussian bandwidth. Events within a trajectory can be dependent.

TABLE II: Pairing-offset sensitivity and cumulative-interaction references at $T = 3000$. Primed factors retain boundary singletons after shifting the pairing. $R_T = C_T D_T = C'_T D'_T$, up to rounding. All nine $Q_T > 1$, including the seven conditions with negative cumulative adjacent interaction.

Model	η	C_T	D_T	C'_T	D'_T	$R_T^{\perp}$	Q_T
MLP	0.004	0.9999	0.7942	1.0000	0.7942	0.0273	848.35
MLP	0.02	0.0983	0.4961	0.0992	0.4914	0.0244	3.99
MLP	0.04	0.1139	0.2254	0.1144	0.2244	0.0199	1.66
ResNet	0.025	0.2026	0.6836	0.2029	0.6829	0.0222	38.80
ResNet	0.05	0.1592	0.3870	0.1601	0.3849	0.0213	8.38
ResNet	0.1	0.1558	0.2399	0.1571	0.2380	0.0202	3.43
Transformer	0.05	0.9999	0.7224	0.9999	0.7224	0.0290	621.73
Transformer	0.1	0.2000	0.6976	0.2004	0.6963	0.0195	51.11
Transformer	0.2	0.0859	0.6300	0.0854	0.6339	0.0184	8.67

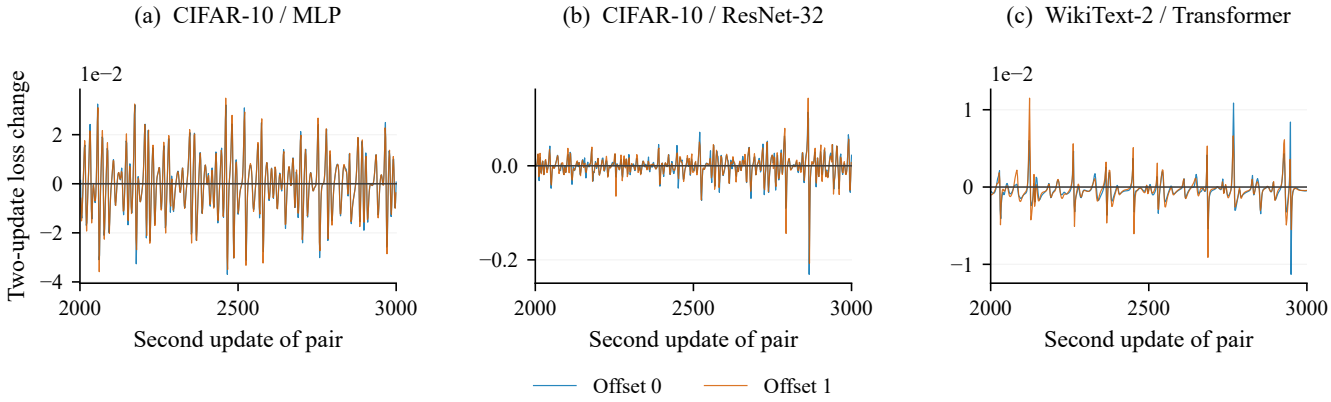


Fig. 10: **Pairing does not ensure local descent.** Every unsmoothed two-update full-training-loss change in the last 1,000 updates at the highest rate per model. Offset 0 contains 500 pairs over states 2000–3000; offset 1 contains 499 interior pairs over states 2001–2999. The horizontal line is zero. Separate vertical scales preserve each model's variation. Each window has negative total change despite positive individual changes.

References

[1] J. Hestness, S. Narang, N. Ardalani, G. Diamos, H. Jun, H. Kianinejad, M. Patwary, Y. Yang, and Y. Zhou, "Deep learning scaling is predictable, empirically," *arXiv preprint arXiv:1712.00409*, 2017.

[2] J. Kaplan, S. McCandlish, T. Henighan *et al.*, "Scaling laws for neural language models," *arXiv preprint arXiv:2001.08361*, 2020.

[3] T. Henighan *et al.*, "Scaling laws for autoregressive generative modeling," *arXiv preprint arXiv:2010.14701*, 2020.

[4] J. Hoffmann, S. Borgeaud, A. Mensch, E. Cassirer, T. Jones, T. Dasilva, K. Tunyasuvunakool, C. Humphreys, E. Rutherford, E. Buchatskaya *et al.*, "Training compute-optimal large language models," *arXiv preprint arXiv:2203.15556*, 2022.

[5] P. Goyal, P. Dollár, R. Girshick, P. Noordhuis, L. Wesolowski, A. Kyrola, A. Tulloch, Y. Jia, and K. He, "Accurate, large minibatch sgd: Training imagenet in 1 hour," *arXiv preprint arXiv:1706.02677*, 2017.

[6] L. N. Smith, "Cyclical learning rates for training neural networks," in *2017 IEEE winter conference on applications of computer vision (WACV)*. IEEE, 2017, pp. 464–472.

[7] I. Loshchilov and F. Hutter, "Sgdr: Stochastic gradient descent with warm restarts," *arXiv preprint arXiv:1608.03983*, 2016.

[8] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *International conference on machine learning*. pmlr, 2015, pp. 448–456.

[9] A. Krogh and J. Hertz, "A simple weight decay can improve generalization," *Advances in neural information processing systems*, vol. 4, 1991.

[10] B. T. Polyak, "Some methods of speeding up the convergence of iteration methods," *USSR Computational Mathematics and Mathematical Physics*, vol. 4, pp. 1–17, 1964.

[11] D. P. Bertsekas, *Nonlinear Programming*, 2nd ed. Belmont, MA: Athena Scientific, 1999.

[12] S. Jastrzębski, M. Szymczak, S. Fort, D. Arpit, J. Tabor, K. Cho, and K. Geras, "The break-even point on optimization trajectories of deep neural networks," in *International Conference on Learning Representations (ICLR)*, 2020.

[13] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar, "Gradient descent on neural networks typically occurs at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2021.

[14] S. Arora, Z. Li, and A. Panigrahi, "Understanding gradient descent on the edge of stability in deep learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 948–1024.

[15] A. Damian, E. Nichani, and J. D. Lee, "Self-stabilization: The implicit bias of gradient descent at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2023.

[16] S. P. Singh, B. He, T. Hofmann, and B. Schölkopf, "The directionality of optimization trajectories in neural networks," in *International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/37771cc0be272368102a37f202bb88d8-Abstract-Conference.html

[17] L. Sagun, U. Evci, V. U. Guney, Y. Dauphin, and L. Bottou, "Empirical analysis of the hessian of over-parametrized neural networks," 2017.

[18] B. Ghorbani, S. Krishnan, and Y. Xiao, "An investigation into neural net optimization via hessian eigenvalue density," in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 2232–2241.

[19] S. Mandt, M. D. Hoffman, and D. M. Blei, "Stochastic gradient descent as approximate bayesian inference," *Journal of Machine Learning Research*, vol. 18, no. 134, pp. 1–35, 2017. [Online]. Available: https://www.jmlr.org/papers/v18/17-214.html

[20] W. Hu, C. J. Li, L. Li, and J.-G. Liu, "On the diffusion approximation of nonconvex stochastic gradient descent," *Annals of Mathematical Sciences and Applications*, vol. 4, no. 1, pp. 3–32, 2019.

[21] S. Jastrzębski, Z. Kenton, D. Arpit, N. Ballas, A. Fischer, Y. Bengio, and A. Storkey, "Three factors influencing minima in sgd," *arXiv preprint arXiv:1711.04623*, 2017. [Online]. Available: https://arxiv.org/abs/1711.04623

[22] J. M. Cohen, A. Damian, A. Talwalkar, J. Z. Kolter, and J. D. Lee, "Understanding optimization in deep learning with central flows," in *International Conference on Learning Representations*, 2025.

[23] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts *et al.*, "PaLM: Scaling language modeling with pathways," *arXiv preprint arXiv:2204.02311*, 2022.

[24] S. Zhang, S. Roller, N. Goyal, M. Artetxe, M. Chen, S. Chen *et al.*, "OPT: Open pre-trained transformer language models," *arXiv preprint arXiv:2205.01068*, 2022.

[25] Team OLMo, P. Walsh, L. Soldaini, D. Groeneveld, K. Lo *et al.*, "2 OLMo 2 furious," *arXiv preprint arXiv:2501.00656*, 2025.

[26] Kimi Team *et al.*, "Kimi K2: Open agentic intelligence," *arXiv preprint arXiv:2507.20534*, 2025.

[27] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[28] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training recurrent neural networks," in *International conference on machine learning*. Pmlr, 2013, pp. 1310–1318.

[29] J. Zhang, T. He, S. Sra, and A. Jadbabaie, "Why gradient clipping accelerates training: A theoretical justification for adaptivity," in *International Conference on Learning Representations*, 2020. [Online]. Available: https://openreview.net/forum?id=BJgnXpVYwS

[30] PyTorch contributors, "PyTorch examples: Word language model," GitHub repository, revision `13acec6d7c78`; full commit specified in the URL. [Online]. Available: https://github.com/pytorch/examples/tree/13acec6d7c78dacd5e1fe9b0b4a325e1d39abc15/word_language_model